

STRUCTURAL ISOMERISM

A Comprehensive, Publication-Quality Textbook for JEE Main, JEE Advanced & Chemistry Olympiads

PEDAGOGICAL BOUNDARY NOTICE: STRICT EXCLUSION OF TAUTOMERISM

In accordance with rigorous chemical taxonomy and modern examination guidelines, tautomerism is completely excluded from this treatise. Dynamic equilibria involving rapid proton transfer (e.g., keto–enol, imine–enamine, nitro–aci-nitro, lactam–lactim, amide–imidic acid, ring–chain tautomerism, and annular tautomerism) represent distinct chemical equilibria with low activation energies. Pairs representing tautomers are not treated as structural isomers in this text. Stable constitutional isomers that do not interconvert spontaneously under ambient conditions (such as allylic alcohols vs. saturated aldehydes, or enol ethers) are included and explicitly annotated.

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SECTION 1: INTRODUCTION TO ISOMERISM

The phenomenon of isomerism lies at the very heart of structural organic chemistry. First recognized in the 1820s by Justus von Liebig and Friedrich Wöhler—when silver cyanate (AgNCO) and silver fulminate (AgCNO) were shown to share an identical elemental composition yet exhibit vastly disparate physical and chemical properties—isomerism demonstrated that molecular identity is governed not merely by the elemental inventory of a compound, but crucially by the specific spatial and connectivity arrangements of its constituent atoms. In 1830, Jöns Jacob Berzelius coined the term 'isomerism' (from the Greek *isos*, 'equal', and *meros*, 'part') to formalize this principle.

In modern competitive examinations such as JEE Main and JEE Advanced, structural isomerism forms the bedrock upon which mechanistic organic chemistry, retrosynthetic analysis, and spectroscopic structure elucidation are built. A mastery of isomerism demands the ability to systematically enumerate all viable topological graphs for a given molecular formula without duplication, omit chemically impossible structures (such as those violating Bredt's rule or tetrahedral valency limits), and distinguish true structural differences from simple conformational rotations or stereoisomeric configurations.

SECTION 2: MOLECULAR FORMULA VS STRUCTURAL FORMULA

A rigorous analytical workflow begins by drawing a sharp line between empirical, molecular, and structural formulas:

- 1. Empirical Formula:** The simplest integer ratio of atoms of each element in a compound. For example, glucose ($\text{C}_6\text{H}_{12}\text{O}_6$), acetic acid ($\text{C}_2\text{H}_4\text{O}_2$), and formaldehyde (CH_2O) all share the identical empirical formula CH_2O . Empirical formulas alone provide zero structural information and cannot be used to deduce isomerism.
- 2. Molecular Formula:** The actual absolute number of atoms of each element comprising one discrete molecule. Identical molecular formula is the absolute, non-negotiable prerequisite for two species to be isomers. If two compounds differ in molecular formula by even a single hydrogen atom or oxygen atom, they are not isomers (they may be homologues, oxidation products, or unrelated compounds).
- 3. Structural (Constitutional) Formula:** The explicit representation of the sequence of covalent attachments between atomic nuclei. A structural formula can be expressed as an expanded Lewis structure, a condensed linear formula (e.g., $\text{CH}_3\text{CH}_2\text{OH}$ vs. CH_3OCH_3), or a bond-line (skeletal) graph. It specifies precisely which atom is bonded to which, unambiguously resolving structural ambiguity.

SECTION 3: CONSTITUTION / CONNECTIVITY

In formal IUPAC terminology, the constitution of a chemical species refers to the complete specification of the sequence of chemical bonds (single, double, or triple) between atoms in a molecule, without regard to their three-dimensional arrangement in space. The atomic connectivity represents the topological adjacency graph where vertices denote atomic nuclei and edges represent covalent bonds.

Two molecules are constitutional (structural) isomers if and only if they possess the identical molecular formula but non-isomorphic connectivity graphs. Graph isomorphism dictates that if no one-to-one mapping exists between the vertices of graph G_1 and graph G_2 that preserves bond orders and atom types, the two compounds are distinct constitutional isomers. In contrast, if an isomorphism exists, the two drawings represent the identical compound (or a stereoisomeric pair).

SECTION 4: CONSTITUTIONAL VS STEREOISOMERISM

The division between constitutional isomerism and stereoisomerism is fundamental to all competitive examination taxonomies:

- **Constitutional (Structural) Isomers:** Molecules that possess different atom-to-atom connectivity. For instance, butan-1-ol and diethyl ether (both $\text{C}_4\text{H}_{10}\text{O}$) have different atoms attached to the oxygen atom. They differ in physical properties (boiling point, melting point, density) and exhibit completely distinct chemical reactivity.
- **Stereoisomers:** Molecules that possess the exact same connectivity (the identical sequence of bonded atoms), but differ exclusively in the three-dimensional spatial orientation of their atoms. Stereoisomers are partitioned into: (a) Configurational isomers (enantiomers, diastereomers, cis/trans, E/Z), which cannot be interconverted without breaking formal covalent bonds; and (b) Conformational isomers (rotamers), which interconvert spontaneously via simple rotation about single sigma bonds (e.g., anti vs. gauche butane).

CRITICAL JEE EXAMINATION RULE ON QUESTION STEM TERMINOLOGY

Whenever a JEE Main or JEE Advanced question asks for the 'number of structural isomers' or 'number of constitutional isomers', STEREOISOMERS MUST NOT BE COUNTED SEPARATELY. Enantiomeric pairs (d/l) and geometric isomers (cis/trans or E/Z) represent ONE structural isomer. Conversely, if the question stem asks for 'total number of isomers (including stereoisomers)', both stereoisomers and structural isomers must be counted.

SECTION 5: DECISION TREE FOR ISOMER CLASSIFICATION

To eliminate ambiguity when classifying any pair of chemical species in competitive exams, apply the following deterministic decision tree:

Algorithmic Step	Diagnostic Decision Logic
Step 1: Check Molecular Formula	Do the two species have the exact same molecular formula? If NO -> NOT ISOMERS (e.g., homologues, different oxidation states). If YES -> Proceed to Step 2.
Step 2: Check Atom Connectivity	Do the two species have the identical sequence of atomic bonds? If YES -> STEREOISOMERS (Configurational / Conformational). If NO -> STRUCTURAL ISOMERS. Proceed to Step 3.
Step 3: Check Ring Presence	Is one isomer an open-chain compound and the other a cyclic compound? If YES -> RING-CHAIN ISOMERS. If NO -> Proceed to Step 4.
Step 4: Check Functional Groups	Do the two species belong to different functional group classes (e.g., alcohol vs. ether, aldehyde vs. ketone, acid vs. ester, nitrile vs. isonitrile, 1° vs. 2° amine)? If YES -> FUNCTIONAL-GROUP ISOMERS. If NO -> Proceed to Step 5.
Step 5: Check Polyvalent Heteroatoms	Do the species possess the same polyvalent functional group (-O-, -S-, -CO-, -COO-, -NH-) with different alkyl chain distributions attached to either side? If YES -> METAMERS. If NO -> Proceed to Step 6.
Step 6: Check Principal Carbon Skeleton	Do the compounds have different parent carbon chains (different length, branching, or ring size)? If YES -> CHAIN / SKELETAL ISOMERS. If NO -> Proceed to Step 7.
Step 7: Check Locants / Substituent Positions	Do the compounds have the identical parent chain and identical functional groups, differing solely in the locant numbers of substituents or multiple bonds? If YES -> POSITION ISOMERS.

SECTION 6: DEGREE OF UNSATURATION (DBE / IHD)

The Degree of Unsaturation, also designated as the Double Bond Equivalent (DBE) or Index of Hydrogen Deficiency (IHD), represents the total number of pi bonds and/or rings present in a molecule. Calculating DBE is the mandatory, non-negotiable first step in solving any isomer counting problem.

First-Principles Derivation: For an alkane containing C carbon atoms, the saturated hydrogen count is $2C + 2$. Every introduction of a pi bond or a closed ring reduces the hydrogen count by exactly 2. For a general organic molecule containing carbon (C), hydrogen (H), nitrogen (N), halogens (X = F, Cl, Br, I), and divalent atoms (oxygen O, sulfur S):

THE UNIVERSAL HETEROATOM DBE EQUATION

$$\text{DBE} = C + 1 - (H + X - N) / 2$$

- Group 14 (Tetravalent, C, Si): Contribute +1 to baseline unsaturation.
- Group 15 (Trivalent, N, P): Each nitrogen increases the hydrogen capacity by 1 (subtract N/2).
- Group 16 (Divalent, O, S): Divalent atoms form two bonds and insert between C-H or C-C bonds without altering the required hydrogen count (O and S are omitted from the calculation).
- Group 17 (Monovalent, X = F, Cl, Br, I): Each halogen replaces one hydrogen atom (add X to H).

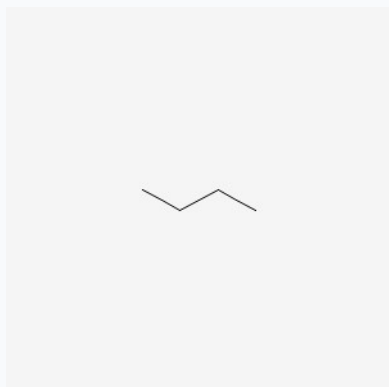
DBE	Structural Possibilities	Pi Bonds	Rings	Representative Formulas
0	Acyclic saturated alkane, alcohol, ether, or amine	0	0	C ₄ H ₁₀ (Butane), C ₄ H ₁₀ O (Butanol)
1	One double bond (alkene, carbonyl C=O, imine C=N) OR one monocyclic ring (cycloalkane, cyclic ether)	1 or 0	0 or 1	C ₄ H ₈ (Butene / Cyclobutane), C ₃ H ₆ O (Propanal / Oxetane)
2	One triple bond (alkyne, nitrile C#N), OR two double bonds (diene, allene), OR one double bond + one ring (cycloalkene), OR two rings (bicycloalkane, spiroalkane)	2, 1, or 0	0, 1, or 2	C ₄ H ₆ (But-1-yne, Butadiene, Cyclobutene, Bicyclobutane)
4	Benzene ring (aromatic sextet: 1 six-membered ring + 3 formal conjugated pi bonds)	3	1	C ₆ H ₆ (Benzene), C ₇ H ₈ O (Cresols, Benzyl alcohol)
5	Monosubstituted benzene with one double bond (e.g., styrene, benzaldehyde, benzoic acid)	4	1	C ₈ H ₈ (Styrene), C ₇ H ₆ O (Benzaldehyde)

SECTION 7: CHAIN / SKELETAL ISOMERISM

Chain (skeletal) isomerism occurs when two or more compounds have the identical molecular formula and identical functional groups, but possess different carbon skeletons. The difference arises from variation in the length of the principal (parent) carbon chain, the presence and nature of alkyl branches, or the size of a carbon ring.

Diagnostic Identification Rule: To diagnose chain isomerism: Determine the IUPAC name of both compounds. If the parent root word changes (e.g., butane vs. propane, pentane vs. butane) while the principal functional groups remain identical, the compounds are chain isomers.

Example EX-001 — C₄H₁₀ (Chain Isomerism, DBE = 0)



Structure A: Butane

PubChem CID: 7843 | SMILES: CCCC

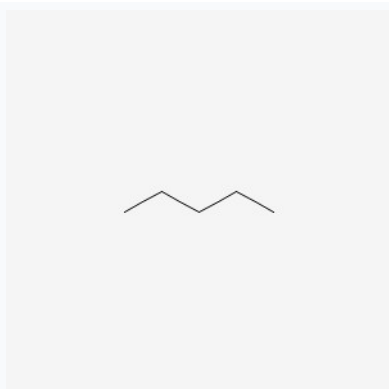


Structure B: 2-Methylpropane

PubChem CID: 6360 | SMILES: CC(C)C

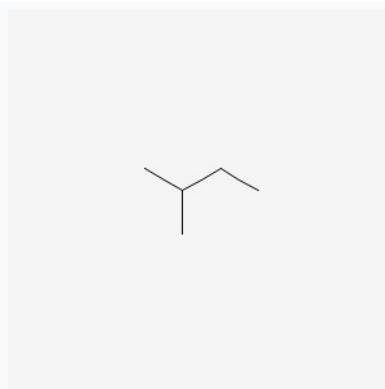
- **Classification:** Chain Isomerism
- **Exact Connectivity Difference:** Unbranched C₄ main chain vs branched C₃ chain with methyl at C₂
- **Formula & Valency Verification:** Identical formula C₄H₁₀, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Fundamental alkane chain pair; butane has a 4-carbon chain while 2-methylpropane has a 3-carbon chain.

Example EX-002 — C₅H₁₂ (Chain Isomerism, DBE = 0)



Structure A: Pentane

PubChem CID: 8003 | SMILES: CCCCC



Structure B: 2-Methylbutane

PubChem CID: 6556 | SMILES: CCC(C)C

- **Classification:** Chain Isomerism
- **Exact Connectivity Difference:** Linear C₅ chain vs branched C₄ chain with methyl substituent
- **Formula & Valency Verification:** Identical formula C₅H₁₂, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Linear C₅ chain vs branched C₄ chain with a single methyl substituent at C₂.

Example EX-003 — C₅H₁₂ (Chain Isomerism, DBE = 0)



Structure A: Pentane

PubChem CID: 8003 | SMILES: CCCCC



Structure B: 2,2-Dimethylpropane

PubChem CID: 10 041 | SMILES: CC(C)(C)C

- **Classification:** Chain Isomerism
- **Exact Connectivity Difference:** Linear C₅ chain vs quaternary C₃ chain with two methyl groups
- **Formula & Valency Verification:** Identical formula C₅H₁₂, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Linear C₅ chain vs quaternary C₃ skeleton with two geminal methyl groups (neopentane).

Example EX-016 — C₄H₈ (Chain Isomerism, DBE = 1)



Structure A: But-1-ene

PubChem CID: 7844 | SMILES: C=CCC

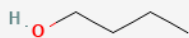


Structure B: 2-Methylprop-1-ene

PubChem CID: 8255 | SMILES: C=C(C)C

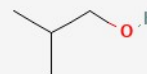
- **Classification:** Chain Isomerism
- **Exact Connectivity Difference:** Unbranched C₄ chain with terminal alkene vs C₃ chain branched at C₂
- **Formula & Valency Verification:** Identical formula C₄H₈, DBE = 1. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Terminal straight-chain but-1-ene vs branched isobutene (2-methylprop-1-ene).

Example EX-035 — C₄H₁₀O (Chain Isomerism, DBE = 0)



Structure A: Butan-1-ol

PubChem CID: 263 | SMILES: CCCCCO

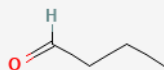


Structure B: 2-Methylpropan-1-ol

PubChem CID: 6560 | SMILES: CC(C)CO

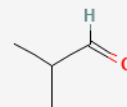
- **Classification:** Chain Isomerism
- **Exact Connectivity Difference:** Linear C₄ primary alcohol vs branched isobutyl alcohol
- **Formula & Valency Verification:** Identical formula C₄H₁₀O, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Primary alcohol on linear C₄ chain vs primary alcohol on branched isobutyl C₃ skeleton.

Example EX-040 — C₄H₈O (Chain Isomerism, DBE = 1)



Structure A: Butanal

PubChem CID: 261 | SMILES: CCCC=O

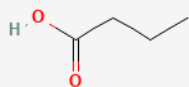


Structure B: 2-Methylpropanal

PubChem CID: 6561 | SMILES: CC(C)C=O

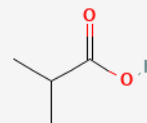
- **Classification:** Chain Isomerism
- **Exact Connectivity Difference:** Linear C₄ aldehyde vs C₃ branched aldehyde (isobutyraldehyde)
- **Formula & Valency Verification:** Identical formula C₄H₈O, DBE = 1. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Straight-chain butanal vs branched 2-methylpropanal (isobutyraldehyde).

Example EX-044 — C₄H₈O₂ (Chain Isomerism, DBE = 1)



Structure A: Butanoic acid

PubChem CID: 264 | SMILES: CCCC(=O)O



Structure B: 2-Methylpropanoic acid

PubChem CID: 6590 | SMILES: CC(C)C(=O)O

- **Classification:** Chain Isomerism
- **Exact Connectivity Difference:** Linear C₄ butyric acid vs branched isobutyric acid
- **Formula & Valency Verification:** Identical formula C₄H₈O₂, DBE = 1. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Unbranched butanoic acid vs branched 2-methylpropanoic acid.

SECTION 8: POSITION ISOMERISM

Position isomerism occurs when two or more compounds have the identical molecular formula, identical carbon skeleton, and identical functional groups, but differ solely in the position (locant) of a functional group, substituent, or multiple bond on the carbon chain or ring.

Diagnostic Identification Rule: Determine the IUPAC parent root word. If both compounds share the identical parent chain root and identical substituents/functional groups, differing exclusively in their numerical locants (e.g., butan-1-ol vs. butan-2-ol, or ortho vs. meta vs. para in aromatic systems), they are position isomers.

Example EX-051 — C₄H₈ (Position Isomerism, DBE = 1)



Structure A: But-1-ene

PubChem CID: 7844 | SMILES: C=CCC

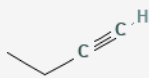


Structure B: But-2-ene

PubChem CID: 12220 | SMILES: CC=CC

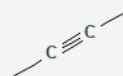
- **Classification:** Position Isomerism
- **Exact Connectivity Difference:** Terminal double bond (C₁=C₂) vs internal double bond (C₂=C₃)
- **Formula & Valency Verification:** Identical formula C₄H₈, DBE = 1. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Alkene position isomerism: double bond at terminal C₁ vs internal C₂ on identical C₄ chain.

Example EX-058 — C₄H₆ (Position Isomerism, DBE = 2)



Structure A: But-1-yne

PubChem CID: 7846 | SMILES: C#CCC



Structure B: But-2-yne

PubChem CID: 10419 | SMILES: CC#CC

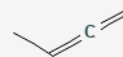
- **Classification:** Position Isomerism
- **Exact Connectivity Difference:** Terminal alkyne (C1) vs internal alkyne (C2) on C₄ chain
- **Formula & Valency Verification:** Identical formula C₄H₆, DBE = 2. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Alkyne position isomerism: terminal but-1-yne vs internal but-2-yne on identical C₄ chain.

Example EX-063 — C₄H₆ (Position Isomerism, DBE = 2)



Structure A: Buta-1,3-diene

PubChem CID: 7845 | SMILES: C=CC=C

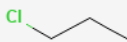


Structure B: Buta-1,2-diene

PubChem CID: 11535 | SMILES: C=C=CC

- **Classification:** Position Isomerism
- **Exact Connectivity Difference:** Conjugated double bonds (1,3) vs cumulative allene double bonds (1,2)
- **Formula & Valency Verification:** Identical formula C₄H₆, DBE = 2. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Diene position isomerism: conjugated buta-1,3-diene vs cumulated allene buta-1,2-diene.

Example EX-066 — C₃H₇Cl (Position Isomerism, DBE = 0)



Structure A: 1-Chloropropane

PubChem CID: 10 899 | SMILES: CCCC1

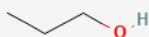


Structure B: 2-Chloropropane

PubChem CID: 6361 | SMILES: CC(C)Cl

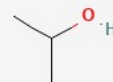
- **Classification:** Position Isomerism
- **Exact Connectivity Difference:** Chlorine at C1 (primary) vs C2 (secondary) on propane chain
- **Formula & Valency Verification:** Identical formula C₃H₇Cl, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Haloalkane position isomerism: 1-chloropropane (primary) vs 2-chloropropane (secondary).

Example EX-078 — C₃H₈O (Position Isomerism, DBE = 0)



Structure A: Propan-1-ol

PubChem CID: 1031 | SMILES: CCCO

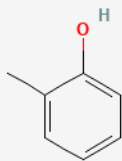


Structure B: Propan-2-ol

PubChem CID: 3776 | SMILES: CC(C)O

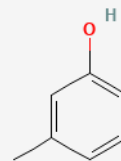
- **Classification:** Position Isomerism
- **Exact Connectivity Difference:** Hydroxyl group at C1 (primary) vs C2 (secondary) on propane chain
- **Formula & Valency Verification:** Identical formula C₃H₈O, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Alcohol position isomerism: propan-1-ol (primary) vs propan-2-ol (secondary).

Example EX-087 — C₇H₈O (Position Isomerism, DBE = 4)



Structure A: 2-Methylphenol

PubChem CID: 335 | SMILES: Cc1ccccc1O

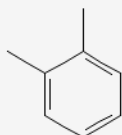


Structure B: 3-Methylphenol

PubChem CID: 342 | SMILES: Cc1cccc(O)c1

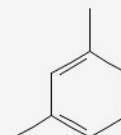
- **Classification:** Position Isomerism
- **Exact Connectivity Difference:** ortho (1,2) vs meta (1,3) relationship of methyl and hydroxyl on benzene
- **Formula & Valency Verification:** Identical formula C₇H₈O, DBE = 4. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Aromatic position isomerism: ortho-cresol (1,2-locants) vs meta-cresol (1,3-locants).

Example EX-090 — C₈H₁₀ (Position Isomerism, DBE = 4)



Structure A: 1,2-Dimethylbenzene

PubChem CID: 7237 | SMILES: Cc1ccccc1C



Structure B: 1,3-Dimethylbenzene

PubChem CID: 7929 | SMILES: Cc1cccc(C)c1

- **Classification:** Position Isomerism
- **Exact Connectivity Difference:** ortho (1,2) vs meta (1,3) positions of two methyl groups on benzene
- **Formula & Valency Verification:** Identical formula C₈H₁₀, DBE = 4. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Aromatic dialkyl position isomerism: ortho-xylene (1,2) vs meta-xylene (1,3).

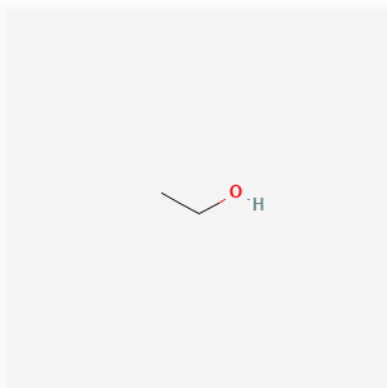
SECTION 9: FUNCTIONAL-GROUP ISOMERISM

Functional-group isomerism occurs when compounds possessing the identical molecular formula belong to different homologous series and contain different functional groups. Because different functional groups possess distinct electronic distributions, bond energies, and reaction mechanisms, functional isomers exhibit vastly different physical and chemical behaviors.

Formula	DBE	Class A	Class B	Example Compound A	Example Compound B
C _n H _{2n+2} O	0	Monohydric Alcohols	Ethers	Ethanol (CH ₃ CH ₂ OH)	Methoxymethane (CH ₃ OCH ₃)

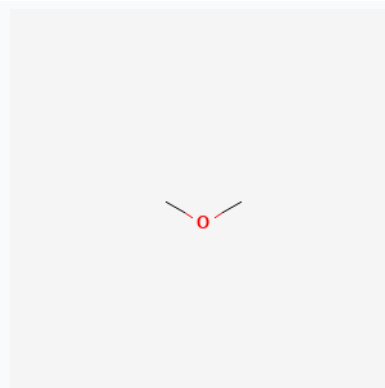
Formula	DBE	Class A	Class B	Example Compound A	Example Compound B
C _n H _{2n} O	1	Aldehydes	Ketones	Propanal (CH ₃ CH ₂ CHO)	Propan-2-one (CH ₃ COCH ₃)
C _n H _{2n} O	1	Unsaturated Alcohols (Alkenols)	Cyclic Ethers (Oxiranes)	Prop-2-en-1-ol (allyl alc)	Oxetane / Methyloxirane
C _n H _{2n} O ₂	1	Carboxylic Acids	Esters	Ethanoic acid (CH ₃ COOH)	Methyl formate (HCOOCH ₃)
C _n H _{2n} O ₂	1	Hydroxy-carbonyls	Dicarbonyls / Cyclic acetals	1-Hydroxypropan-2-one	Propanoic acid derivatives
C _n H _{2n} +3N	0	Primary (1°) Amines	Secondary (2°) & Tertiary (3°) Amines	Propan-1-amine (R-NH ₂)	N-Methylethanamine (R ₂ NH)
C _n H _{2n} -1N	2	Nitriles (Alkyl cyanides)	Isonitriles (Alkyl isocyanides)	Acetonitrile (CH ₃ -C#N)	Methyl isocyanide (CH ₃ -N#C)
C _n H _{2n} +1NO ₂	1	Nitroalkanes (R-NO ₂)	Alkyl Nitrites (R-O-N=O)	Nitroethane (CH ₃ CH ₂ NO ₂)	Ethyl nitrite (CH ₃ CH ₂ ONO)

Example EX-093 — C₂H₆O (Functional Isomerism, DBE = 0)



Structure A: Ethanol

PubChem CID: 702 | SMILES: CCO

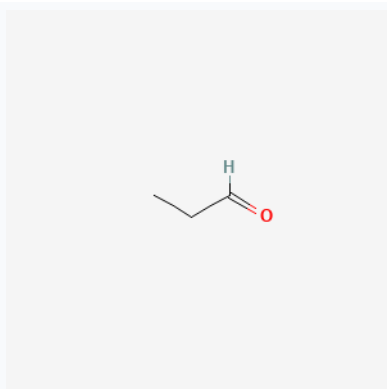


Structure B: Methoxymethane

PubChem CID: 8254 | SMILES: COC

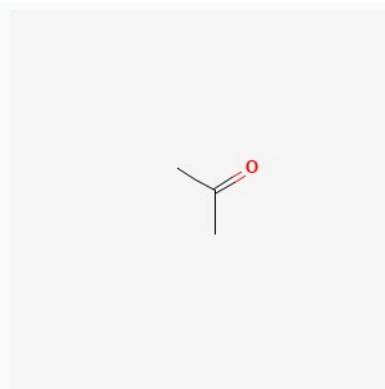
- **Classification:** Functional Isomerism
- **Exact Connectivity Difference:** Primary alcohol (-OH) vs dialkyl ether (-O-)
- **Formula & Valency Verification:** Identical formula C₂H₆O, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Classic alcohol vs ether pair: ethanol (polar, H-bonding) vs dimethyl ether (gas at room temp).

Example EX-099 — C₃H₆O (Functional Isomerism, DBE = 1)



Structure A: Propanal

PubChem CID: 527 | SMILES: CCC=O



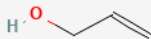
Structure B: Propan-2-one

PubChem CID: 180 | SMILES: CC(C)=O

- **Classification:** Functional Isomerism
- **Exact Connectivity Difference:** Aldehyde (-CH=O) vs ketone (-C(=O)-)
- **Formula & Valency Verification:** Identical formula C₃H₆O, DBE = 1. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Carbonyl functional isomerism: propanal (oxidizable with

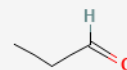
Tollens/Fehling) vs propan-2-one (ketone).

Example EX-105 — C₃H₆O (Functional Isomerism, DBE = 1)



Structure A: Prop-2-en-1-ol

PubChem CID: 7858 | SMILES: C=CCO

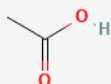


Structure B: Propanal

PubChem CID: 527 | SMILES: CCC=O

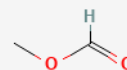
- **Classification:** Functional Isomerism
- **Exact Connectivity Difference:** Allylic unsaturated alcohol (stable, non-tautomeric C=C-C-OH) vs saturated aldehyde
- **Formula & Valency Verification:** Identical formula C₃H₆O, DBE = 1. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Unsaturated alcohol vs saturated aldehyde: allyl alcohol vs propanal.

Example EX-110 — C₂H₄O₂ (Functional Isomerism, DBE = 1)



Structure A: Ethanoic acid

PubChem CID: 176 | SMILES: CC(=O)O



Structure B: Methyl formate

PubChem CID: 7865 | SMILES: COC=O

- **Classification:** Functional Isomerism
- **Exact Connectivity Difference:** Carboxylic acid (-COOH) vs formate ester (-COOCH₃)
- **Formula & Valency Verification:** Identical formula C₂H₄O₂, DBE = 1. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Acid vs ester functional isomerism: ethanoic acid vs methyl formate.

Example EX-121 — C₂H₃N (Functional Isomerism, DBE = 2)



Structure A: Acetonitrile

PubChem CID: 6342 | SMILES: CC#N

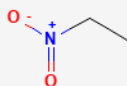


Structure B: Methyl isocyanide

PubChem CID: 11 646 | SMILES: [C-]#[N+]C

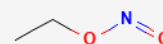
- **Classification:** Functional Isomerism
- **Exact Connectivity Difference:** Nitrile (-C#N, carbon-linked) vs isonitrile (-N#C, nitrogen-linked)
- **Formula & Valency Verification:** Identical formula C₂H₃N, DBE = 2. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Nitrile vs isonitrile functional isomerism: acetonitrile vs toxic, pungent methyl isocyanide.

Example EX-125 — C₂H₅NO₂ (Functional Isomerism, DBE = 1)



Structure A: Nitroethane

PubChem CID: 6587 | SMILES: CC[N+](=O)[O-]

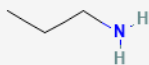


Structure B: Ethyl nitrite

PubChem CID: 8026 | SMILES: CCON=O

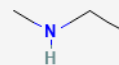
- **Classification:** Functional Isomerism
- **Exact Connectivity Difference:** Nitro compound (C-NO₂ bond) vs alkyl nitrite ester (C-O-N=O bond)
- **Formula & Valency Verification:** Identical formula C₂H₅NO₂, DBE = 1. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Nitroalkane vs alkyl nitrite: nitroethane vs ethyl nitrite.

Example EX-128 — C₃H₉N (Functional Isomerism, DBE = 0)



Structure A: Propan-1-amine

PubChem CID: 7852 | SMILES: CCCN

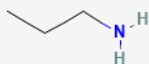


Structure B: N-Methylethanamine

PubChem CID: 12 219 | SMILES: CCNC

- **Classification:** Functional Isomerism
- **Exact Connectivity Difference:** Primary amine (-NH₂) vs secondary amine (-NH-)
- **Formula & Valency Verification:** Identical formula C₃H₉N, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Amine classification: primary amine (propan-1-amine) vs secondary amine (N-methylethanamine).

Example EX-129 — C₃H₉N (Functional Isomerism, DBE = 0)



Structure A: Propan-1-amine

PubChem CID: 7852 | SMILES: CCCN



Structure B: N,N-Dimethylmethanamine

PubChem CID: 1146 | SMILES: CN(C)C

- **Classification:** Functional Isomerism
- **Exact Connectivity Difference:** Primary amine (-NH₂) vs tertiary amine (-N<)
- **Formula & Valency Verification:** Identical formula C₃H₉N, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Amine classification: primary amine (propan-1-amine) vs tertiary amine (trimethylamine).

SECTION 10: METAMERISM

Metamerism is a specialized form of structural isomerism exhibited by compounds containing a polyvalent (multivalent) functional group or heteroatom bridging two or more alkyl residues. Metamers possess the identical functional group and identical molecular formula, but differ in the nature or size of the alkyl groups attached to either side of the bridging polyvalent group.

Qualifying Polyvalent Groups: Ethers (-O-), Thioethers (-S-), Ketones (-CO-), Esters (-COO-), Secondary Amines (-NH-), Tertiary Amines (-N<), Amides (-CONH-, -CON<), Acid Anhydrides (-CO-O-CO-), and Sulfoxides/Sulfones (-SO-, -SO₂-). Monovalent functional groups (-OH, -CHO, -COOH, -CN, -X) CANNOT exhibit metamerism because they possess only one attachment point.

Minimum Carbon Requirements: • Ethers require at least $C_4H_{10}O$ (e.g., diethyl ether vs. methyl propyl ether).

- Ketones require at least $C_5H_{10}O$ (e.g., pentan-2-one vs. pentan-3-one).
- Esters require at least $C_4H_8O_2$ (e.g., ethyl acetate vs. methyl propanoate vs. propyl formate).
- Secondary amines require at least $C_4H_{11}N$ (e.g., diethylamine vs. methylpropylamine).

Example EX-141 — $C_5H_{12}O$ (Metamerism Isomerism, DBE = 0)



Structure A: 1-Methoxybutane

PubChem CID: 12 338 | SMILES: CCCOC



Structure B: 1-Ethoxypropane

PubChem CID: 12 340 | SMILES: CCCOCC

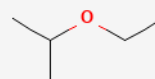
- **Classification:** Metamerism Isomerism
- **Exact Connectivity Difference:** Methyl + Butyl (1+4) vs Ethyl + Propyl (2+3) around ether oxygen
- **Formula & Valency Verification:** Identical formula $C_5H_{12}O$, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Ether metamerism: symmetrical diethyl ether (2+2) vs unsymmetrical methyl propyl ether (1+3).

Example EX-142 — $C_5H_{12}O$ (Metamerism Isomerism, DBE = 0)



Structure A: 1-Methoxybutane

PubChem CID: 12 338 | SMILES: CCCOC

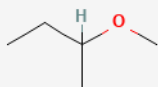


Structure B: 2-Ethoxypropane

PubChem CID: 12 256 | SMILES: CCOC(C)C

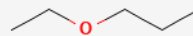
- **Classification:** Metamerism Isomerism
- **Exact Connectivity Difference:** Methyl + Butyl (1+4) vs Ethyl + Isopropyl (2+3) around ether oxygen
- **Formula & Valency Verification:** Identical formula $C_5H_{12}O$, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Ether metamerism: diethyl ether (2+2) vs methyl isopropyl ether (1+3 with branched alkyl).

Example EX-143 — C₅H₁₂O (Metamerism Isomerism, DBE = 0)



Structure A: 2-Methoxybutane

PubChem CID: 23 238 | SMILES: CCC(C)OC

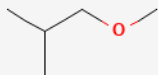


Structure B: 1-Ethoxypropane

PubChem CID: 12 340 | SMILES: CCCOCC

- **Classification:** Metamerism Isomerism
- **Exact Connectivity Difference:** Methyl + sec-Butyl (1+4) vs Ethyl + Propyl (2+3) around ether oxygen
- **Formula & Valency Verification:** Identical formula C₅H₁₂O, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Ketone metamerism: pentan-2-one (1+3) vs symmetrical pentan-3-one (2+2).

Example EX-144 — C₅H₁₂O (Metamerism Isomerism, DBE = 0)



Structure A: 1-Methoxy-2-methylpropane

PubChem CID: 12 250 | SMILES: COCC(C)C



Structure B: 1-Ethoxypropane

PubChem CID: 12 340 | SMILES: CCCOCC

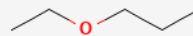
- **Classification:** Metamerism Isomerism
- **Exact Connectivity Difference:** Methyl + Isobutyl (1+4) vs Ethyl + Propyl (2+3) around ether oxygen
- **Formula & Valency Verification:** Identical formula C₅H₁₂O, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Ester metamerism: ethyl propanoate vs propyl acetate across the bridging ester group.

Example EX-145 — C₅H₁₂O (Metamerism Isomerism, DBE = 0)



Structure A: 2-Methoxy-2-methylpropane

PubChem CID: 15 413 | SMILES: COC(C)(C)C



Structure B: 1-Ethoxypropane

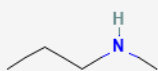
PubChem CID: 12 340 | SMILES: CCCOCC

- **Classification:** Metamerism Isomerism
- **Exact Connectivity Difference:** Methyl + tert-Butyl (1+4) vs Ethyl + Propyl (2+3) around ether oxygen
- **Formula & Valency Verification:** Identical formula C₅H₁₂O, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Secondary amine metamerism: diethylamine (2+2) vs methylpropylamine (1+3) across -NH-.

SECTION 11: RING-CHAIN ISOMERISM

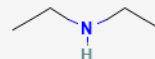
Ring-chain isomerism occurs when one structural isomer possesses an open-chain (acyclic) structure with one or more multiple bonds, while the other isomer possesses a closed-ring (cyclic) structure. This phenomenon arises because the closure of an open carbon chain into a ring requires the formal elimination of two terminal hydrogen atoms, exactly matching the degree of unsaturation (DBE = 1) introduced by a double bond.

Example EX-151 — C₄H₁₁N (Metamerism Isomerism, DBE = 0)



Structure A: N-Methylpropan-1-amine

PubChem CID: 12 315 | SMILES: CCCNC

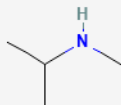


Structure B: Diethylamine

PubChem CID: 8021 | SMILES: CCNCC

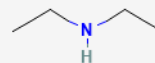
- **Classification:** Metamerism Isomerism
- **Exact Connectivity Difference:** Methyl + Propyl (1+3) vs Ethyl + Ethyl (2+2) around secondary amine -NH-
- **Formula & Valency Verification:** Identical formula C₄H₁₁N, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Aliphatic prop-1-ene (DBE=1, 1 double bond) vs cyclopropane (DBE=1, 1 carbocyclic ring).

Example EX-152 — C₄H₁₁N (Metamerism Isomerism, DBE = 0)



Structure A: N-Methylpropan-2-amine

PubChem CID: 78 485 | SMILES: CNC(C)C

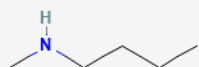


Structure B: Diethylamine

PubChem CID: 8021 | SMILES: CCNCC

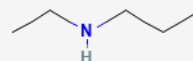
- **Classification:** Metamerism Isomerism
- **Exact Connectivity Difference:** Methyl + Isopropyl (1+3) vs Ethyl + Ethyl (2+2) around -NH-
- **Formula & Valency Verification:** Identical formula C₄H₁₁N, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Linear but-1-ene vs four-membered cyclobutane carbocycle.

Example EX-153 — C₅H₁₃N (Metamerism Isomerism, DBE = 0)



Structure A: N-Methylbutan-1-amine

PubChem CID: 8068 | SMILES: CCCCNC

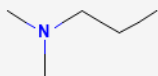


Structure B: N-Ethylpropan-1-amine

PubChem CID: 88 398 | SMILES: CCCNCC

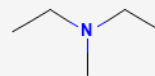
- **Classification:** Metamerism Isomerism
- **Exact Connectivity Difference:** Methyl + Butyl (1+4) vs Ethyl + Propyl (2+3) around -NH-
- **Formula & Valency Verification:** Identical formula C₅H₁₃N, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Linear but-1-ene vs substituted methylcyclopropane carbocycle.

Example EX-156 — C₅H₁₃N (Metamerism Isomerism, DBE = 0)



Structure A: N,N-Dimethylpropan-1-amine

PubChem CID: 61 236 | SMILES: CCCN(C)C

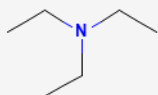


Structure B: N-Ethylethanamine

PubChem CID: 12 022 | SMILES: CCN(C)CC

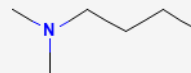
- **Classification:** Metamerism Isomerism
- **Exact Connectivity Difference:** Methyl + Methyl + Propyl (1,1,3) vs Ethyl + Ethyl + Methyl (2,2,1) around tertiary nitrogen
- **Formula & Valency Verification:** Identical formula C₅H₁₃N, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Alkyne but-1-yne (DBE=2, 1 triple bond) vs cyclobutene (DBE=2, 1 ring + 1 double bond).

Example EX-157 — C₆H₁₅N (Metamerism Isomerism, DBE = 0)



Structure A: Triethylamine

PubChem CID: 8471 | SMILES: CCN(CC)CC

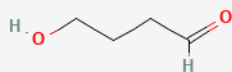


Structure B: N,N-Dimethylbutan-1-amine

PubChem CID: 13 563 | SMILES: CCCCN(C)C

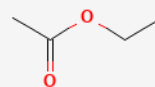
- **Classification:** Metamerism Isomerism
- **Exact Connectivity Difference:** Ethyl + Ethyl + Ethyl (2,2,2) vs Methyl + Methyl + Butyl (1,1,4) around tertiary nitrogen
- **Formula & Valency Verification:** Identical formula C₆H₁₅N, DBE = 0. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Conjugated buta-1,3-diene (DBE=2, 2 double bonds) vs cyclobutene (DBE=2, 1 ring + 1 double bond).

Example EX-120 — C₄H₈O₂ (Functional Isomerism, DBE = 1)



Structure A: 4-Hydroxybutanal

PubChem CID: 93 093 | SMILES: O=CCCCO



Structure B: Ethyl acetate

PubChem CID: 8857 | SMILES: CCOC(C)=O

- **Classification:** Functional Isomerism
- **Exact Connectivity Difference:** gamma-Hydroxyaldehyde vs ester
- **Formula & Valency Verification:** Identical formula C₄H₈O₂, DBE = 1. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Cyclohexene (monocyclic alkene, DBE=2) vs fused bicyclo[2.2.0]hexane (bicyclic alkane, DBE=2).

Example EX-121 — C₂H₃N (Functional Isomerism, DBE = 2)



Structure A: Acetonitrile

PubChem CID: 6342 | SMILES: CC#N



Structure B: Methyl isocyanide

PubChem CID: 11 646 | SMILES: [C-]#[N+]C

- **Classification:** Functional Isomerism
- **Exact Connectivity Difference:** Nitrile (-C#N, carbon-linked) vs isonitrile (-N#C, nitrogen-linked)
- **Formula & Valency Verification:** Identical formula C₂H₃N, DBE = 2. Distinct non-isomorphic constitutional graphs.
- **Common JEE Trap / Examination Note:** Cyclohexene (monocyclic alkene, DBE=2) vs fused bicyclo[3.1.0]hexane (bicyclic alkane, DBE=2).

SECTION 12: CLASSIFICATION OVERLAPS & THE JEE PRIORITY RULE

In competitive examinations, two compounds frequently display characteristics of more than one structural isomer class. For example, diethyl ether and methyl isopropyl ether differ in the alkyl groups on either side of the oxygen (metamers), but their alkyl groups also differ in carbon chain skeleton (chain isomers). When answering single-choice objective questions, students must follow the standardized Indian competitive examination priority protocol:

THE MASTER JEE STRUCTURAL ISOMERISM HIERARCHY RULE

Ring-Chain Isomerism > Functional-Group Isomerism > Metamerism > Chain Isomerism > Position Isomerism

1. Ring-Chain vs. Functional: An open-chain alkene and a cyclic alkane are strictly Ring-Chain isomers (preferred over Functional).
2. Functional vs. Metamerism: If functional groups differ (e.g., alcohol vs. ether, or ester vs. carboxylic acid), the pair is Functional.
3. Metamerism vs. Chain / Position: For compounds containing the same polyvalent functional group with different alkyl distributions, Metamerism takes absolute priority over Chain or Position isomerism.
4. Chain vs. Position: If two compounds differ in both the parent chain skeleton and substituent locants, Chain isomerism takes priority.

SECTION 13: SYMMETRY AND EQUIVALENT POSITIONS

The number of monoderivatives (e.g., monochloro, monohydroxy) that a hydrocarbon can form is determined strictly by its molecular symmetry and the number of chemically non-equivalent sets of hydrogen atoms. In graph theory, non-equivalent positions correspond to the distinct orbits under the automorphism group of the molecular graph.

- Propane ($\text{CH}_3\text{-CH}_2\text{-CH}_3$): 2 sets of equivalent hydrogens (1° methyl H, 2° methylene H) $\rightarrow$ 2 monochloro derivatives.
- Butane ($\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$): 2 sets of equivalent hydrogens (C_1 methyl, C_2 methylene) $\rightarrow$ 2 monochloro derivatives.
- 2-Methylpropane [$\text{CH}(\text{CH}_3)_3$]: 2 sets of equivalent hydrogens (9 primary H, 1 tertiary methine H) $\rightarrow$ 2 monochloro derivatives.
- Pentane ($\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-CH}_3$): 3 sets of equivalent hydrogens (C_1 , C_2 , C_3) $\rightarrow$ 3 monochloro derivatives.
- 2-Methylbutane: 4 sets of equivalent hydrogens (C_1 , C_2 , C_3 , C_4) $\rightarrow$ 4 monochloro derivatives.
- 2,2-Dimethylpropane (Neopentane): All 12 hydrogens are chemically identical $\rightarrow$ Exactly 1 monochloro derivative.

SECTION 14: SYSTEMATIC ENUMERATION METHODS

To achieve a 100% success rate in JEE counting problems without missing isomers or introducing duplicates, follow the Three-Tier Exhaustive Algorithm:

Step 1: Calculate the DBE from the molecular formula to determine the required combination of rings and multiple bonds.

Step 2: Partition the DBE into discrete structural sub-families (e.g., for DBE = 1: acyclic alkenes vs. monocarbocycles).

Step 3: For acyclic systems, enumerate the parent carbon skeletons in descending order of main-chain length (n, n-1 with methyl, n-2 with dimethyl or ethyl).

Step 4: For each carbon skeleton, identify all symmetry-independent positions and insert the functional group or multiple bond.

Step 5: For cyclic systems, enumerate rings in descending order of ring size (6-membered, 5-membered with methyl, 4-membered with ethyl or dimethyl, 3-membered with propyl, isopropyl, ethyl-methyl, or trimethyl).

Step 6: Screen every candidate graph for valency violations, Bredt's rule violations, and topological equivalence.

SECTION 15: ALKANE ISOMER COUNTING (C_4H_{10} TO $\text{C}_{10}\text{H}_{22}$)

Alkanes ($\text{C}_n\text{H}_{2n+2}$, DBE = 0) exhibit purely chain (skeletal) isomerism. The number of constitutional isomers grows rapidly with carbon number:

Formula	Constitutional Isomers	Exhaustive Constitutional Isomer Names
C_4H_{10}	2	Butane, 2-Methylpropane (Isobutane)
C_5H_{12}	3	Pentane, 2-Methylbutane (Isopentane), 2,2-Dimethylpropane (Neopentane)
C_6H_{14}	5	Hexane, 2-Methylpentane, 3-Methylpentane, 2,2-Dimethylbutane, 2,3-Dimethylbutane
C_7H_{16}	9	Heptane, 2-Methylhexane, 3-Methylhexane, 2,2-Dimethylpentane, 2,3-Dimethylpentane, 2,4-Dimethylpentane, 3,3-Dimethylpentane, 3-Ethylpentane, 2,2,3-Trimethylbutane
C_8H_{18}	18	Octane (1), Methylheptanes (3), Dimethylhexanes (6), Ethylhexanes (1), Trimethylpentanes (4), Ethylmethylpentanes (2), Tetramethylbutane (1)
C_9H_{20}	35	Nonane (1), Methyloctanes (4), Dimethylheptanes (10), Ethylheptanes (2), etc.

Formula	Constitutional Isomers	Exhaustive Constitutional Isomer Names
C10H22	75	75 constitutional isomers (standard benchmark in Cayley graph theory)

SECTION 16: ALKENE ISOMER COUNTING (C4H8 & C5H10)

Alkenes and cycloalkanes have DBE = 1. The total structural isomer count is divided into acyclic alkenes and carbocycles:

- C₄H₈ (Total ₅ structural isomers): ₃ acyclic alkenes (But-₁-ene, But-₂-ene, ₂-Methylprop-₁-ene) + ₂ carbocycles (Cyclobutane, Methylcyclopropane).
- C₅H₁₀ (Total ₁₀ structural isomers): ₅ acyclic alkenes (Pent-₁-ene, Pent-₂-ene, ₂-Methylbut-₁-ene, ₂-Methylbut-₂-ene, ₃-Methylbut-₁-ene) + ₅ carbocycles (Cyclopentane, Methylcyclobutane, Ethylcyclopropane, _{1,1}-Dimethylcyclopropane, _{1,2}-Dimethylcyclopropane).

SECTION 17: ALKYNE AND DIENE CASES (C4H6 & C5H8)

With DBE = 2, C₄H₆ yields exactly ₉ constitutional isomers across four structural partitions:

- ₁. Alkynes (₂ isomers): But-₁-yne, But-₂-yne.
- ₂. Dienes & Allenes (₂ isomers): Buta-_{1,3}-diene, Buta-_{1,2}-diene.
- ₃. Monocyclic Alkenes (₂ isomers): Cyclobutene, ₁-Methylcyclopropene, ₃-Methylcyclopropene, Methylenecyclopropane.
- ₄. Bicyclic Alkanes (₁ isomer): Bicyclo[_{1,1,0}]butane.

SECTION 18: ALCOHOL AND ETHER ISOMERISM

For saturated monohydric alcohols and ethers (C_nH_{2n+2}O, DBE = 0):

- C₃H₈O: Exactly ₃ constitutional isomers (₂ alcohols: Propan-₁-ol, Propan-₂-ol; ₁ ether: Methoxyethane).
- C₄H₁₀O: Exactly ₇ constitutional isomers (₄ alcohols: Butan-₁-ol, Butan-₂-ol, ₂-Methylpropan-₁-ol, ₂-Methylpropan-₂-ol; ₃ ethers: Diethyl ether, ₁-Methoxypropane, ₂-Methoxypropane).
- C₅H₁₂O: Exactly ₁₄ constitutional isomers (₈ alcohols: Pentan-₁-ol, Pentan-₂-ol, Pentan-₃-ol, ₂-Methylbutan-₁-ol, ₃-Methylbutan-₁-ol, ₂-Methylbutan-₂-ol, ₃-Methylbutan-₂-ol, _{2,2}-Dimethylpropan-₁-ol; ₆ ethers: Methyl butyl ether, Methyl sec-butyl ether, Methyl isobutyl ether, Methyl tert-butyl ether, Ethyl propyl ether, Ethyl isopropyl ether).

SECTION 19: ALDEHYDE AND KETONE ISOMERISM

For carbonyl compounds (C_nH_{2n}O, DBE = 1, carbonyl only):

- C₃H₆O: Exactly ₂ carbonyl isomers (₁ aldehyde: Propanal; ₁ ketone: Propan-₂-one).
- C₄H₈O: Exactly ₃ carbonyl isomers (₂ aldehydes: Butanal, ₂-Methylpropanal; ₁ ketone: Butan-₂-one).
- C₅H₁₀O: Exactly ₇ carbonyl isomers (₄ aldehydes: Pentanal, ₂-Methylbutanal, ₃-Methylbutanal, _{2,2}-Dimethylpropanal; ₃ ketones: Pentan-₂-one, Pentan-₃-one, ₃-Methylbutan-₂-one).

SECTION 20: CARBOXYLIC ACID AND ESTER ISOMERISM

For carboxylic acids and esters (C_nH_{2n}O₂, DBE = ₁):

- C₃H₆O₂: Exactly ₃ isomers (₁ acid: Propanoic acid; ₂ esters: Methyl acetate, Ethyl formate).
- C₄H₈O₂: Exactly ₆ isomers (₂ acids: Butanoic acid, ₂-Methylpropanoic acid; ₄ esters: Ethyl acetate, Methyl propanoate, Propyl formate, Isopropyl formate).
- C₅H₁₀O₂: Exactly ₁₃ isomers (₄ acids: Pentanoic acid, ₂-Methylbutanoic acid, ₃-Methylbutanoic acid, _{2,2}-Dimethylpropanoic acid; ₉ esters: Methyl butanoate, Methyl ₂-methylpropanoate, Ethyl propanoate, Propyl acetate, Isopropyl acetate, Butyl formate, sec-Butyl formate, Isobutyl formate, tert-Butyl formate).

SECTION 21: AMINE ISOMERISM (C3H9N & C4H11N)

Amines (C_nH_{2n+3}N, DBE = 0) partition into primary (1°, -NH₂), secondary (2°, -NH-), and tertiary (3°, -N<) classes, which represent functional isomers:

- C₃H₉N (₄ isomers): Two ₁° amines (Propan-₁-amine, Propan-₂-amine) + One ₂° amine (N-Methylethanamine) + One ₃° amine (N,N-Dimethylmethanamine).
- C₄H₁₁N (₈ isomers): Four ₁° amines (Butan-₁-amine, Butan-₂-amine, ₂-Methylpropan-₁-amine, ₂-Methylpropan-₂-amine) + Three ₂° amines (Diethylamine, N-Methylpropan-₁-amine, N-Methylpropan-₂-amine) + One ₃° amine (N,N-Dimethylethanamine).

SECTION 22: NITRILE AND ISONITRILE ISOMERISM

Nitriles ($\text{-C}\equiv\text{N}$) and isonitriles ($\text{-N}\equiv\text{C}$) have DBE = 2 and represent functional isomers:

- $\text{C}_2\text{H}_3\text{N}$: Acetonitrile ($\text{CH}_3\text{-C}\equiv\text{N}$) vs. Methyl isocyanide ($\text{CH}_3\text{-[N}^+\text{]\# [C}^-\text{]}$).
- $\text{C}_3\text{H}_5\text{N}$: Propanenitrile ($\text{CH}_3\text{CH}_2\text{-C}\equiv\text{N}$) vs. Ethyl isocyanide ($\text{CH}_3\text{CH}_2\text{-[N}^+\text{]\# [C}^-\text{]}$).

SECTION 23: HALO COMPOUND ISOMERISM

Halogenated alkanes exhibit extensive position and chain isomerism:

- $\text{C}_3\text{H}_7\text{Cl}$ (2 isomers): 1-Chloropropane, 2-Chloropropane.
- $\text{C}_4\text{H}_9\text{Cl}$ (4 isomers): 1-Chlorobutane, 2-Chlorobutane, 1-Chloro-2-methylpropane, 2-Chloro-2-methylpropane.
- $\text{C}_3\text{H}_6\text{Cl}_2$ (4 isomers): 1,1-Dichloropropane (geminal), 2,2-Dichloropropane (geminal), 1,2-Dichloropropane (vicinal), 1,3-Dichloropropane (isolated).
- $\text{C}_3\text{H}_5\text{Cl}_3$ (5 isomers): 1,1,1-Trichloropropane, 1,1,2-Trichloropropane, 1,1,3-Trichloropropane, 1,2,2-Trichloropropane, 1,2,3-Trichloropropane.

SECTION 24: CYCLIC SYSTEMS (MONOCYCLIC, FUSED, BRIDGED, SPIRO)

Cyclic constitutional isomers include diverse topological ring architectures:

- Monocyclic systems: Cyclohexane vs. Methylcyclopentane vs. Ethylcyclobutane vs. Propylcyclopropane (C_6H_{12} , DBE=1).
- Fused bicyclic systems: Bicyclo[2.2.0]hexane (two cyclobutanes sharing an edge), Bicyclo[3.1.0]hexane (5-ring fused to 3-ring).
- Bridged bicyclic systems: Bicyclo[2.2.1]heptane (Norbornane), Bicyclo[1.1.1]pentane.
- Spirocyclic systems: Spiropentane (two 3-rings sharing one quaternary carbon), Spiro[2.3]hexane.

SECTION 25: AROMATIC STRUCTURAL ISOMERISM

Aromatic structural isomerism (DBE ≥ 4) is governed by ring substitution locants and side-chain partitioning:

- $\text{C}_7\text{H}_8\text{O}$ (5 aromatic isomers): Three isomeric cresols (2-Methylphenol, 3-Methylphenol, 4-Methylphenol) + Benzyl alcohol + Anisole (Methoxybenzene).
- C_8H_{10} (4 aromatic hydrocarbons): Three isomeric xylenes (1,2-Dimethylbenzene, 1,3-Dimethylbenzene, 1,4-Dimethylbenzene) + Ethylbenzene.
- $\text{C}_7\text{H}_9\text{N}$ (5 aromatic amines): Three isomeric toluidines (2-Methylaniline, 3-Methylaniline, 4-Methylaniline) + N-Methylaniline + Benzylamine.

SECTION 26: POLYFUNCTIONAL COMPOUNDS & COMPETING LOCANTS

When a molecule contains multiple functional groups, isomerism arises from both chain branching and relative functional group locants. For example, hydroxy-ketones vs. carboxylic acids ($\text{C}_4\text{H}_8\text{O}_2$), amino-ketones ($\text{C}_4\text{H}_9\text{NO}$: 3-aminobutan-2-one vs. 4-aminobutan-2-one), and keto-esters exhibit complex structural interconversions while strictly adhering to atomic valency and IUPAC numbering rules.

SECTION 27: COMMON JEE EXAMINATION TRAPS & PITFALLS

- **Trap 1: Counting Stereoisomers When Structural Isomers Are Asked:** When the question stem specifies 'structural isomers', enantiomeric pairs and geometric cis/trans isomers must NOT be counted separately. But-2-ene is ONE structural isomer, not two.
- **Trap 2: Accidentally Including Dynamic Tautomers:** Keto and enol forms (e.g., acetone vs. prop-1-en-2-ol) are tautomers in rapid dynamic equilibrium. They must be excluded from ordinary constitutional isomer counts unless specifically requested.
- **Trap 3: Counting Symmetrical Drawings Twice:** In linear chains, drawing a substituent from left-to-right vs. right-to-left (e.g., 2-chloropentane drawn from either end) represents the exact same compound.
- **Trap 4: Violating Bredt's Rule at Bridgehead Carbons:** A double bond cannot terminate at a bridgehead carbon in small bicyclic systems (e.g., bicyclo[2.2.1]hept-1-ene is impossible under ordinary conditions due to geometric ring strain).
- **Trap 5: Drawing Impossible Ring Sizes with Multiple Bonds:** A triple bond cannot exist in a 3, 4, 5, or 6-membered carbocycle due to severe linear 180° bond angle strain (cyclooctyne is the smallest isolable cycloalkyne).
- **Trap 6: Misclassifying Chain vs. Position in Substituted Skeletons:** 1-Chloropentane vs. 2-chloro-2-methylbutane are CHAIN isomers because their parent carbon chain lengths differ (pentane vs. butane).
- **Trap 7: Forgetting Zero-DBE Oxygen Classes:** Given $\text{C}_4\text{H}_{10}\text{O}$, students often count the 4 alcohols and forget the 3 ethers (or vice versa). Always partition by functional group class first.

- **Trap 8: The Hidden Isocyanide / Isonitrile Family:** Given C_2H_3N , students often write acetonitrile (CH_3CN) and forget methyl isocyanide (CH_3NC). Both have $DBE = 2$ and are constitutional isomers.
- **Trap 9: Alkoxy Branching vs. Chain Isomerism in Metamers:** Diethyl ether vs. methyl isopropyl ether are METAMERS, not chain isomers, because Metamerism takes precedence across polyvalent heteroatoms.
- **Trap 10: Symmetrical Poly-Substituted Benzene Duplication:** In 1,2,4-trimethylbenzene vs. 1,3,4-trimethylbenzene, numbering from different positions can lead students to count the same isomer twice.

SECTION 28: JEE MAIN SOLVED EXAMPLES

The following 10 solved concept drill problems illustrate standard single-concept applications in JEE Main:

Problem 1: Find the total number of structural isomers for C₄H₁₀.

Solution: DBE = 4 + 1 - (10/2) = 0. Acyclic alkanes only. C₄ skeleton has 2 isomers: Butane and 2-Methylpropane. Answer: 2.

Problem 2: What is the total number of constitutional isomers with formula C₃H₈O?

Solution: DBE = 0. Alcohols (C_nH_{2n+1}OH) and Ethers (R-O-R'). Alcohols: Propan-1-ol, Propan-2-ol (2). Ethers: Methoxyethane (1). Total = 3. Answer: 3.

Problem 3: Classify the isomeric relationship between Pentan-2-one and Pentan-3-one.

Solution: Pentan-2-one and pentan-3-one have the identical functional group (-CO-) with different alkyl groups on either side (CH₃/C₂H₅ vs C₂H₅/C₂H₅). By the JEE priority rule, this is Metamerism. Answer: Metamerism.

Problem 4: How many acyclic carbonyl structural isomers exist for C₃H₆O?

Solution: DBE = 3 + 1 - (6/2) = 1. Acyclic carbonyls: Propanal (aldehyde, 1) and Propan-2-one (ketone, 1). Total = 2. Answer: 2.

Problem 5: How many monochloro derivatives are formed upon photochemical chlorination of propane?

Solution: Propane has 2 sets of equivalent H atoms (6 primary, 2 secondary). Thus, 2 monochloro derivatives: 1-chloropropane and 2-chloropropane. Answer: 2.

SECTION 29: JEE ADVANCED SOLVED EXAMPLES

The following multi-concept problems reflect the depth and rigor of JEE Advanced examinations:

Problem 1 (Multi-Concept Challenge):

An organic hydrocarbon (A) with molecular formula C₅H₈ reacts with excess H₂ in the presence of Pd/C to form cyclopentane (B).

Upon treatment with ammoniacal AgNO₃, compound (A) gives NO precipitate. How many constitutional isomers are possible for (A)?

Solution:

1. Molecular formula C₅H₈ has DBE = 5 + 1 - (8/2) = 2.
2. Hydrogenation yields cyclopentane, proving that (A) contains a five-membered carbocyclic ring.
3. Since DBE = 2 and the ring accounts for 1 unit of unsaturation, the remaining unsaturation must be a double bond inside or outside the ring.
4. Cyclopentene is C₅H₈ (1 double bond inside a 5-membered ring). No terminal alkyne exists (consistent with negative Tollens test).
5. Thus, exactly 1 constitutional isomer (Cyclopentene) satisfies all conditions. Answer: 1.

SECTION 30: SINGLE CORRECT QUESTIONS (LEVEL 1 DRILL)

Q1. The isomeric pair Diethyl ether and Methyl propyl ether represents:

(A) Chain isomers (B) Position isomers (C) Metamers (D) Functional isomers. [Correct: C]

Q2. The total number of structural isomers of monochlorobutane (C₄H₉Cl) is:

(A) 2 (B) 3 (C) 4 (D) 5. [Correct: C]

Q3. Propan-1-amine and N-Methylethanamine are related as:

(A) Metamerism (B) Position isomerism (C) Chain isomerism (D) Functional-group isomerism. [Correct: D]

Q4. The total number of structural isomers for C₄H₈ (including rings) is:

(A) 3 (B) 4 (C) 5 (D) 6. [Correct: C]

SECTION 31: MULTIPLE CORRECT QUESTIONS

Q1. Which of the following statements regarding structural isomerism are CORRECT?

- (A) Two species with different empirical formulas cannot be constitutional isomers.
(B) Metamerism takes priority over chain isomerism for compounds with polyvalent heteroatoms.
(C) Stereoisomers have non-isomorphic atomic connectivity graphs.
(D) Prop-1-ene and cyclopropane are ring-chain isomers.

Answer: (A), (B), (D). [Statement C is false: stereoisomers have identical connectivity graphs.]

SECTION 32: NUMERICAL ANSWER / INTEGER QUESTIONS

Q1. Find the total number of primary, secondary, and tertiary alcohols possible for C₄H₁₀O:

Answer: 4. (Butan-1-ol, Butan-2-ol, 2-Methylpropan-1-ol, 2-Methylpropan-2-ol).

Q2. Find the total number of aromatic structural isomers of formula C₈H₁₀:

Answer: 4. (Three isomeric xylenes: ortho, meta, para + Ethylbenzene).

Q3. Find the total number of structural isomers for the alkane C₆H₁₄:

Answer: 5. (Hexane, 2-methylpentane, 3-methylpentane, 2,2-dimethylbutane, 2,3-dimethylbutane).

SECTION 33: MATRIX MATCH QUESTIONS

Match Column I (Isomeric Pair) with Column II (Correct Isomeric Relationship):

Column I: (A) Ethanol & Methoxymethane, (B) But-1-ene & But-2-ene, (C) Butane & 2-Methylpropane, (D) Pentan-2-one & Pentan-3-one.

Column II: (p) Position Isomerism, (q) Chain Isomerism, (r) Functional Isomerism, (s) Metamerism.

Answer: A → r; B → p; C → q; D → s.

SECTION 34: PARAGRAPH-BASED PROBLEMS

Comprehension Passage: A non-cyclic organic compound (X) of molecular formula C₄H₈O₂ gives brisk effervescence with aqueous NaHCO₃. A structural isomer (Y) of the same formula does not react with NaHCO₃ but undergoes alkaline hydrolysis to yield a primary alcohol and an acid salt.

Q₁. Compound (X) exhibits how many constitutional isomers?

Solution: Effervescence with NaHCO₃ confirms a carboxylic acid (-COOH). For C₄H₈O₂, exactly 2 carboxylic acids exist: Butanoic acid and 2-methylpropanoic acid. Answer: 2.

Q₂. How many ester constitutional isomers are possible for compound (Y)?

Solution: C₄H₈O₂ esters: Methyl propanoate, Ethyl acetate, Propyl formate, Isopropyl formate. Total = 4. Answer: 4.

SECTION 35: ADVANCED COUNTING PROBLEMS (OLYMPIAD LEVEL)

Olympiad Challenge: Determine the total number of aromatic structural isomers of formula C₇H₇Cl.

Solution: DBE = 7 + 1 - (7 + 1)/2 = 4. Aromatic ring accounts for all 4 units of unsaturation.

Class 1: Monochlorotoluenes (Chlorine on benzene ring + methyl on benzene ring): 3 isomers (ortho-chlorotoluene, meta-chlorotoluene, para-chlorotoluene).

Class 2: Side-chain chlorinated toluene (Chlorine on methyl group): 1 isomer (Benzyl chloride, C₆H₅CH₂Cl).

Total aromatic structural isomers = 3 + 1 = 4. Answer: 4.

SECTION 36: FORMULA-WISE STRUCTURAL ISOMER INDEX

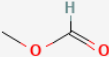
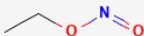
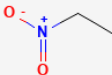
Formula	DBE	Isomer Count	Classes Present	Database References
C4H10	0	2	Alkanes (Chain)	EX-001
C5H12	0	3	Alkanes (Chain)	EX-002, EX-003
C6H14	0	5	Alkanes (Chain)	EX-005 to EX-008
C7H16	0	9	Alkanes (Chain)	EX-011
C4H8	1	5	Alkenes (3) + Carbocycles (2)	EX-016, EX-051, EX-152
C5H10	1	10	Alkenes (5) + Carbocycles (5)	EX-052, EX-082, EX-083
C4H6	2	9	Alkynes (2) + Dienes (2) + Rings (5)	EX-058, EX-063, EX-133
C3H8O	0	3	Alcohols (2) + Ethers (1)	EX-078, EX-094
C4H10O	0	7	Alcohols (4) + Ethers (3)	EX-035, EX-079, EX-141
C3H6O	1	9	Aldehydes (1) + Ketones (1) + Enols/Epoxides (7)	EX-099, EX-105, EX-107
C4H8O	1	3 (Carbonyl)	Aldehydes (2) + Ketones (1)	EX-040, EX-100
C3H6O2	1	3	Carboxylic acids (1) + Esters (2)	EX-111, EX-117
C4H8O2	1	6	Carboxylic acids (2) + Esters (4)	EX-044, EX-144
C3H9N	0	4	1° (2), 2° (1), 3° (1) Amines	EX-084, EX-128, EX-129
C4H11N	0	8	1° (4), 2° (3), 3° (1) Amines	EX-048, EX-145
C7H8O	4	5	Phenols (3), Benzyl alcohol (1), Anisole (1)	EX-087, EX-088
C8H10	4	4	Xylenes (3) + Ethylbenzene (1)	EX-090

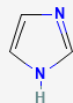
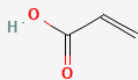
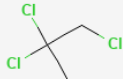
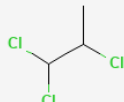
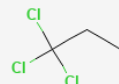
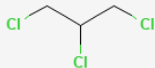
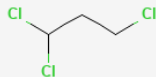
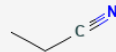
SECTION 37: CATEGORY-WISE STRUCTURAL ISOMER INDEX

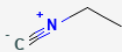
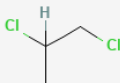
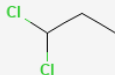
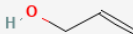
Category	Total Records	Representative Classes	Diagnostic Criterion
Chain / Skeletal Isomerism	102 records	Alkanes, Alkenes, Alkynes, Alkyl Halides, Alcohols, Acids	Parent carbon chain length or branching differs.
Position Isomerism	118 records	Haloalkanes, Alkenes, Dienes, Alcohols, Aromatic cresols/xylenes	Identical carbon skeleton; differing locants.
Functional-Group Isomerism	104 records	Alcohol/Ether, Aldehyde/Ketone, Acid/Ester, Nitrile/Isonitrile	Different functional group families; distinct chemical reactivity.
Metamerism	58 records	Ethers, Ketones, Esters, Secondary Amines, Tertiary Amines	Identical polyvalent group; different attached alkyl groups.
Ring-Chain Isomerism	42 records	Alkene/Cycloalkane, Alkyne/Cycloalkene, Carbonyl/Cyclic Ether	One open-chain unsaturated isomer vs. one closed carbocycle.

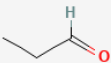
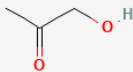
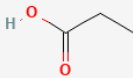
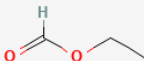
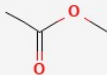
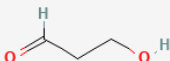
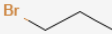
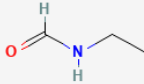
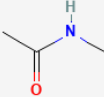
SECTION 38: MOLECULE AND STRUCTURE ATLAS

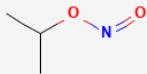
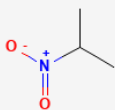
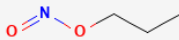
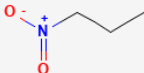
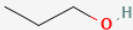
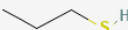
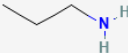
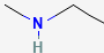
The Molecule and Structure Atlas provides a curated visual gallery of validated 2D chemical structures across foundational organic classes. Every structure depicts verified atomic valency, distinct connectivity, and official NIH PubChem identifiers:

Chemical Structure Gallery (Col 1)	Chemical Structure Gallery (Col 2)	Chemical Structure Gallery (Col 3)
 MOL-001: Acetonitrile C₂H₃N CID: 6342 SMILES: CC#N	 MOL-002: Methyl isocyanide C₂H₃N CID: 11 646 SMILES: [C-]#[N+]C	 MOL-003: 1,1-Dichloroethane C₂H₄Cl₂ CID: 6365 SMILES: CC(Cl)Cl
 MOL-004: 1,2-Dichloroethane C₂H₄Cl₂ CID: 11 SMILES: ClCCCl	 MOL-005: Ethanoic acid C₂H₄O₂ CID: 176 SMILES: CC(=O)O	 MOL-006: Methyl formate C₂H₄O₂ CID: 7865 SMILES: COC=O
 MOL-007: Ethyl nitrite C₂H₅NO₂ CID: 8026 SMILES: CCON=O	 MOL-008: Nitroethane C₂H₅NO₂ CID: 6587 SMILES: CC[N+](=O)[O-]	 MOL-009: Ethanol C₂H₆O CID: 702 SMILES: CCO
 MOL-010: Methoxymethane C₂H₆O CID: 8254 SMILES: COC	 MOL-011: Ethanethiol C₂H₆S CID: 6343 SMILES: CCS	 MOL-012: Dimethyl sulfide C₂H₆S CID: 1068 SMILES: CSC

Chemical Structure Gallery (Col 1)	Chemical Structure Gallery (Col 2)	Chemical Structure Gallery (Col 3)
 MOL-013: Isoxazole C₃H₃NO CID: 9254 SMILES: <chem>c1cnoc1</chem>	 MOL-014: Oxazole C₃H₃NO CID: 9255 SMILES: <chem>c1cocn1</chem>	 MOL-015: 1H-Imidazole C₃H₄N₂ CID: 795 SMILES: <chem>c1c[nH]cn1</chem>
 MOL-016: 1H-Pyrazole C₃H₄N₂ CID: 1048 SMILES: <chem>c1cn[nH]c1</chem>	 MOL-017: Acrylic acid C₃H₄O₂ CID: 6581 SMILES: <chem>C=CC(=O)O</chem>	 MOL-018: Oxetan-2-one C₃H₄O₂ CID: 2365 SMILES: <chem>O=C1CCO1</chem>
 MOL-019: 1,2,2-Trichloropropane C₃H₅Cl₃ CID: 18 505 SMILES: <chem>CC(Cl)(Cl)CCl</chem>	 MOL-020: 1,1,2-Trichloropropane C₃H₅Cl₃ CID: 11 733 SMILES: <chem>CC(Cl)C(Cl)Cl</chem>	 MOL-021: 1,1,1-Trichloropropane C₃H₅Cl₃ CID: 24 622 SMILES: <chem>CCC(Cl)(Cl)Cl</chem>
 MOL-022: 1,2,3-Trichloropropane C₃H₅Cl₃ CID: 7285 SMILES: <chem>ClCC(Cl)CCl</chem>	 MOL-023: 1,1,3-Trichloropropane C₃H₅Cl₃ CID: 33 157 SMILES: <chem>ClCCC(Cl)Cl</chem>	 MOL-024: Propanenitrile C₃H₅N CID: 7854 SMILES: <chem>CCC#N</chem>

Chemical Structure Gallery (Col 1)	Chemical Structure Gallery (Col 2)	Chemical Structure Gallery (Col 3)
 MOL-025: Ethyl isocyanide C₃H₅N CID: 12 226 SMILES: [C-]#[N+]CC	 MOL-026: Cyclopropane C₃H₆ CID: 6351 SMILES: C1CC1	 MOL-027: Prop-1-ene C₃H₆ CID: 8252 SMILES: C=CC
 MOL-028: 2,2-Dichloropropane C₃H₆Cl₂ CID: 11 660 SMILES: CC(C)(Cl)Cl	 MOL-029: 1,2-Dichloropropane C₃H₆Cl₂ CID: 6564 SMILES: CC(Cl)CCl	 MOL-030: 1,1-Dichloropropane C₃H₆Cl₂ CID: 6573 SMILES: CCC(Cl)Cl
 MOL-031: 1,3-Dichloropropane C₃H₆Cl₂ CID: 8881 SMILES: ClCCCCl	 MOL-032: Oxetane C₃H₆O CID: 10 423 SMILES: C1COC1	 MOL-033: Prop-2-en-1-ol C₃H₆O CID: 7858 SMILES: C=CCO
 MOL-034: Methoxyethene C₃H₆O CID: 7861 SMILES: C=COC	 MOL-035: Propan-2-one C₃H₆O CID: 180 SMILES: CC(C)=O	 MOL-036: 2-Methyloxirane C₃H₆O CID: 6378 SMILES: CC1CO1

Chemical Structure Gallery (Col 1)	Chemical Structure Gallery (Col 2)	Chemical Structure Gallery (Col 3)
 MOL-037: Propanal C₃H₆O CID: 527 SMILES: CCC=O	 MOL-038: 1-Hydroxypropan-2-one C₃H₆O₂ CID: 8299 SMILES: CC(=O)CO	 MOL-039: Propanoic acid C₃H₆O₂ CID: 1032 SMILES: CCC(=O)O
 MOL-040: Ethyl formate C₃H₆O₂ CID: 8025 SMILES: CCOC=O	 MOL-041: Methyl acetate C₃H₆O₂ CID: 6584 SMILES: COC(C)=O	 MOL-042: 3-Hydroxypropanal C₃H₆O₂ CID: 75049 SMILES: O=CCCO
 MOL-043: 2-Bromopropane C₃H₇Br CID: 6358 SMILES: CC(C)Br	 MOL-044: 1-Bromopropane C₃H₇Br CID: 7840 SMILES: CCCBr	 MOL-045: 2-Chloropropane C₃H₇Cl CID: 6361 SMILES: CC(C)Cl
 MOL-046: 1-Chloropropane C₃H₇Cl CID: 10899 SMILES: CCCl	 MOL-047: N-Ethylformamide C₃H₇NO CID: 12319 SMILES: CCNC=O	 MOL-048: N-Methylacetamide C₃H₇NO CID: 6582 SMILES: CNC(C)=O

Chemical Structure Gallery (Col 1)	Chemical Structure Gallery (Col 2)	Chemical Structure Gallery (Col 3)
 MOL-049: Isopropyl nitrite C₃H₇NO₂ CID: 10 929 SMILES: CC(C)ON=O	 MOL-050: 2-Nitropropane C₃H₇NO₂ CID: 398 SMILES: CC(C)[N+](=O)[O-]	 MOL-051: Propyl nitrite C₃H₇NO₂ CID: 10 979 SMILES: CCCON=O
 MOL-052: 1-Nitropropane C₃H₇NO₂ CID: 7903 SMILES: CCC[N+](=O)[O-]	 MOL-053: Propan-2-ol C₃H₈O CID: 3776 SMILES: CC(C)O	 MOL-054: Propan-1-ol C₃H₈O CID: 1031 SMILES: CCCO
 MOL-055: Methoxyethane C₃H₈O CID: 10 903 SMILES: CCOC	 MOL-056: Propane-1-thiol C₃H₈S CID: 7848 SMILES: CCCS	 MOL-057: Ethyl methyl sulfide C₃H₈S CID: 12 230 SMILES: CCSC
 MOL-058: Propan-2-amine C₃H₉N CID: 6363 SMILES: CC(C)N	 MOL-059: Propan-1-amine C₃H₉N CID: 7852 SMILES: CCCN	 MOL-060: N-Methylethanamine C₃H₉N CID: 12 219 SMILES: CCNC

SECTION 39: COMPREHENSIVE PRACTICE ASSIGNMENT

- Q₁. Find the total number of cyclic structural isomers possible for the molecular formula C₅H₁₀.
- Q₂. Write all possible structural isomers for the formula C₃H₉N and classify them into 1°, 2°, and 3° amines.
- Q₃. How many monochloro derivatives are formed when 2-methylbutane undergoes photochemical free-radical chlorination?
- Q₄. A compound of formula C₆H₈O gives a positive 2,4-DNP test and negative Fehling's test. Identify the compound and write its IUPAC name.
- Q₅. Classify the relationship between: (a) Methoxyethane and Propan-1-ol, (b) Pentan-2-one and 3-Methylbutan-2-one, (c) But-1-ene and Cyclobutane.
- Q₆. Determine the total number of structural isomers (including cyclic) with formula C₆H₁₂.
- Q₇. How many carboxylic acids and esters are possible for the molecular formula C₄H₈O₂?
- Q₈. An aromatic compound of formula C₇H₆O gives a violet coloration with neutral FeCl₃. How many structural isomers satisfy this condition?
- Q₉. Explain why acetone and prop-1-en-2-ol are not considered ordinary structural isomers in this chapter.
- Q₁₀. Calculate the DBE and total number of acyclic alkynes possible for the formula C₅H₈.

SECTION 40: COMPLETE DETAILED STEP-BY-STEP SOLUTIONS

Solution to Q1: Cyclic Isomers of C₅H₁₀

Formula C₅H₁₀ has DBE = $5 + 1 - (10/2) = 1$. Rings with DBE = 1:

1. 5-membered ring: Cyclopentane (1 isomer).

2. 4-membered ring: Methylcyclobutane (1 isomer).

3. 3-membered ring:

- Mono-substituted: Ethylcyclopropane (1 isomer).

- Di-substituted: 1,1-Dimethylcyclopropane (1 isomer), 1,2-Dimethylcyclopropane (1 structural isomer).

Total cyclic structural isomers = 1 + 1 + 1 + 1 + 1 = 5 isomers. (Note: cis/trans geometric forms of 1,2-dimethylcyclopropane are stereoisomers and count as 1 structural isomer).

Solution to Q2: Amines of C₃H₉N

DBE = $3 + 1 - (9 - 1)/2 = 0$. Saturated aliphatic amines:

- Primary (1°, -NH₂): Propan-1-amine, Propan-2-amine (2 isomers).
- Secondary (2°, -NH-): N-Methylethanamine (1 isomer).
- Tertiary (3°, -N<): N,N-Dimethylmethanamine (1 isomer).

Total constitutional isomers = 4.

Solution to Q3: Monochlorination of 2-Methylbutane

2-Methylbutane: CH₃-CH(CH₃)-CH₂-CH₃. Hydrogen sets:

1. C₁ methyls (two equivalent methyls, 6H) → 1-Chloro-2-methylbutane.
2. C₂ methine (1H) → 2-Chloro-2-methylbutane.
3. C₃ methylene (2H) → 2-Chloro-3-methylbutane.
4. C₄ methyl (3H) → 1-Chloro-3-methylbutane.

All 4 positions are chemically non-equivalent. Exactly 4 structural isomers are formed.

Solution to Q4: Carbonyl Identification

Positive 2,4-DNP confirms a carbonyl group (>C=O). Negative Fehling's test confirms a ketone (aldehydes reduce Fehling's solution, aliphatic ketones do not).

For C₆H₈O, the only ketone is Butan-2-one (CH₃COCH₂CH₃). IUPAC name: Butan-2-one.

Solution to Q5: Isomeric Pair Classifications

- (a) Methoxyethane (ether) and Propan-1-ol (alcohol): Functional-Group Isomers.
- (b) Pentan-2-one (C₅ main chain) and 3-Methylbutan-2-one (C₄ main chain): Chain Isomers.
- (c) But-1-ene (open-chain alkene) and Cyclobutane (4-membered ring): Ring-Chain Isomers.

Solution to Q6: Structural Isomers of C₄H₆

DBE = $4 + 1 - (6/2) = 2$. Structural partitions:

1. Alkynes: But-1-yne, But-2-yne (2).
2. Dienes/Allenes: Buta-1,3-diene, Buta-1,2-diene (2).
3. Cycloalkenes: Cyclobutene, 1-Methylcyclopropene, 3-Methylcyclopropene, Methylenecyclopropane (4).
4. Bicyclics: Bicyclo[1.1.0]butane (1).

Total structural isomers = 9.

Solution to Q7: Carboxylic Acids and Esters of C₄H₈O₂

DBE = 1. Carboxylic acids (2): Butanoic acid, 2-Methylpropanoic acid.

Esters (4): Ethyl acetate, Methyl propanoate, Propyl formate, Isopropyl formate.

Total = 2 + 4 = 6 structural isomers.

Solution to Q8: Phenols of C₇H₈O

Violet coloration with neutral FeCl₃ is the diagnostic test for the phenolic -OH group directly attached to an aromatic ring.

For C₇H₈O, the phenolic isomers are the three isomeric cresols:

1. 2-Methylphenol (ortho-cresol)
2. 3-Methylphenol (meta-cresol)
3. 4-Methylphenol (para-cresol)

Benzyl alcohol (alcohol) and Anisole (ether) do not give FeCl₃ coloration. Exactly 3 isomers satisfy this condition.

Solution to Q9: Tautomer Exclusion Rationale

Acetone (CH₃COCH₃) and prop-1-en-2-ol [CH₂=C(OH)CH₃] represent keto-enol tautomers. Tautomers are constitutional isomers that exist in rapid dynamic equilibrium through spontaneous intramolecular or solvent-mediated proton transfer under ordinary temperature and pressure. Because they interconvert spontaneously, they cannot be isolated as stable constitutional isomers under standard conditions and are treated under Tautomerism rather than static structural isomerism.

Solution to Q10: Acyclic Alkynes of C₅H₈

DBE = 5 + 1 - (8/2) = 2. Acyclic alkynes containing -C≡C-:

1. Straight-chain: Pent-1-yne (CH≡C-CH₂CH₂CH₃) and Pent-2-yne (CH₃-C≡C-CH₂CH₃).
2. Branched: 3-Methylbut-1-yne [CH≡C-CH(CH₃)₂].

(Note: 2-Methylbut-2-yne is impossible because the internal alkyne carbon would be pentavalent).

Total acyclic alkynes = 3.